

Understanding Data

Understanding Data Landscape

The data landscape has evolved significantly in recent years, driven by technological advancements, increasing data sources, and regulatory requirements.

Understanding this landscape is critical; for business, researchers, and analysts to extract meaningful insights and make informed decisions.

Components of the Data Landscape:

- 1.Types of Data
- 2.Data Sources
- 3.Data Collection Methods

Types of Data

Structured Data:

Organized and easily stored in relational databases, including numerical values, customer records, and financial transactions.

Unstructured Data:

Complex and varied formats such as emails, social media posts, images, and videos that require specialized processing techniques.

Semi-Structured Data:

A mix of both structured and unstructured data, such as JSON and XML files, providing flexibility in data storage and retrieval.

Big Data:

Large-scale data sets characterized by volume, velocity, and variety, necessitating advanced analytics and processing tools.

- Big Data involves:
- **Volume**: Massive amounts of data (terabytes to petabytes and beyond)
- **Velocity**: Data flowing in at high speed (e.g., streaming from sensors or social media)
- **Variety**: Different data types (structured, semi-structured, unstructured)
- **Veracity**: Uncertainty or inaccuracy of data
- **Value**: Extracting meaningful insights from the data

Data Sources First-Party Data:

Collected directly from users through interactions with websites, apps, and business platforms.

Second-Party Data:

Shared data obtained from trusted partners for enhanced insights and audience targeting.

Third-Party Data:

Acquired from external providers, commonly used for market research, advertising, and competitive analysis.

Open Data:

Publicly available data from government agencies, research institutions, and organizations to support transparency and innovation.

Type	Description	Source	Example Use Case
First-Party Data	Data you collect directly from your audience/customers	Your own website, app, CRM, surveys, purchase history	Personalizing your website for returning users
Second-Party Data	Someone else's first-party data that is shared with you through a direct partnership	Partner company or publisher	A travel agency shares customer data with a hotel chain
Third-Party Data	Data collected by entities without a direct relationship with the user, aggregated from many sources	Data brokers, ad tech platforms	Targeting ads based on demographic profiles bought from a third-party vendor

Open Data:

Publicly available data from government agencies, research institutions, and organizations to support transparency and innovation.

Manual Data Entry:

Traditional method involving human input, often prone to errors but useful for qualitative insights.

Automated Data Collection:

Leveraging sensors, APIs, web scraping, and IoT devices for efficient and scalable data gathering.

Surveys and Questionnaires:

Direct user feedback collection, valuable for customer experience and market research.

Tracking and Monitoring Systems:

Logs, cookies, and analytics tools used for behavioral analysis and real-time tracking.

Outcomes Data

Definition:

Outcomes data refers to information collected to measure the results or consequences of an action, intervention, or process. It focuses on the end effects, typically reflecting the impact on individuals, groups, organizations, or systems.

- **Purpose:**

- To evaluate effectiveness and success of programs, treatments, projects, or policies.
- To track progress toward specific goals or objectives.
- To inform decision-making and improvements.

- **Examples:**

- In healthcare: patient recovery rates after treatment, quality of life improvements, or reduction in hospital readmissions.
- In education: student graduation rates, test scores, or employment after graduation.
- In business: customer satisfaction scores, sales growth, or market share changes.

- **Key Characteristics:**

- Often quantitative but can be qualitative

Research Data

Definition:

Research data is the raw or processed information collected during scientific or academic investigations. It forms the basis for analysis, interpretation, and conclusions in research studies.

Purpose:

- To generate new knowledge or validate existing theories.
- To answer specific research questions or test hypotheses.
- To document evidence for academic, scientific, or practical inquiry.

Types:

- Quantitative data: numerical data such as statistics, measurements, surveys, experiments.
- Qualitative data: descriptive data such as interview transcripts, observations, texts, images.

Examples:

- Clinical trial results measuring drug efficacy.
- Survey responses about consumer preferences.
- Ethnographic notes from fieldwork.

Key Characteristics:

- Collected systematically following a research design or methodology.

Competitive Data

Definition:

Competitive data consists of information about competitors, market conditions, and industry trends used to analyze the competitive landscape and inform strategic business decisions.

Purpose:

- To understand competitor strengths, weaknesses, and strategies.
- To identify market opportunities and threats.
- To inform pricing, product development, marketing, and positioning.

Examples:

- Competitors' product features, pricing, and market share.
- Customer reviews and feedback on competitors' offerings.
- Industry reports, market trends, and sales data.

Key Characteristics:

- Often gathered from publicly available sources but can include proprietary intelligence.
- Helps anticipate competitor moves and market shifts.

clickstream data (clickstream analytics)

- Clickstream data and clickstream analytics are the processes involved in collecting, analyzing and reporting [aggregate data](#) about which pages a website visitor visits -- and in what order. The path the visitor takes through a website is called the clickstream.
- Clickstreams are categorized into clickstream data and clickstream analytics, which is also referred to as clickstream analysis.
- The clickstream data is the information collected about a user while they browse through a website or use a web browser.
- Clickstream analytics is the process of tracking, analyzing and reporting data on the pages a user visits and user behavior while on a webpage.

Clickstream data (clickstream analytics)

- Websites use clickstream data to show how a user progressed from an initial search or landing page to buying an item or service.
- [Search engines](#) use clickstream data sets to show where a user has searched for a term, when they have clicked on it and if they go back to searching after this. [Internet service providers](#), advertising networks, and IT and telecom organizations also collect clickstream data.

How clickstream data is generated and used



Actions that generate clickstream data

- User login
- Account registration
- Newsletter sign-up
- Search performed
- Product added to cart
- Product purchased
- Feedback provided



What clickstream data is used to analyze

- Ad campaigns
- Marketing campaigns
- Campaign tools
- Product performance
- A/B testing trials

Clickstream data metrics

- Clickstream data includes the following types of [web analytics](#) information:
- whether the individual is a unique or repeat visitor to the website;
- the terms an individual plugs into a search engine;
- what page the individual lands on first;
- the amount of time a user spends on a page;
- the features on the page the user clicks on and engages with;
- when and where an item is added or removed from a cart;
- where the user goes next; and
- when the back button is used.

- Clickstream data collected from a single session of a user interacting with a website may not be useful. However, an organization can use aggregate data gathered from many visitors to improve its website or service.
- For example, if a lot of visitors leave a site after landing on a page with too little information, the organization may need to enhance the page with more valuable information. Likewise, if visitors often land on a page that isn't the website's [homepage](#), then the organization may want to redesign that page to be more inviting and informative to users.
- Clickstream data does not include personal details about a user, and it is typically stored on the server that supports the website. Clickstream data is a [useful addition](#) to data from Google Analytics.

Clickstream analytics and how it is used

- Organizations use clickstream analytics to uncover trends and draw conclusions from different metrics about their websites. This process typically uses a web server [log file](#) to monitor user activity on a website.
- Using the clickstream analysis, an organization can collect data on the number of page visits, views, and unique and repeat visitors. This data provides an idea of how the organization's website performs and it can help approximate the typical user experience ([UX](#)). A website owner can then adjust the site to make it more user-friendly and increase the chance that visitors will stay longer, make a purchase or otherwise interact with the website and the organization behind it.
- Because an extremely large volume of data can be gathered through clickstream analysis, many e-businesses rely on [big data analytics](#) and related tools such as [Hadoop](#) to interpret the data and

Clickstream analysis is effective when used in conjunction with other, more traditional market research, evaluation resources, data sources and strategies.

- There are two levels of clickstream analysis: traffic analytics and [e-commerce](#) analytics.

Traffic analytics

- This analysis operates at the server level. It collects and analyzes the following data sets:
 - how many pages are served to a user;
 - how long it takes each page to load;
 - how often the user hits the browser's back button; and
 - how much data is transmitted before the user moves to a different webpage.

E-commerce analytics

This analysis uses clickstream data to determine the effectiveness of a website in terms of conversions and transactions. It is concerned with the following data points:

- what pages the shopper lingers on;
- what the shopper puts in or takes out of a shopping cart;
- what items the shopper purchases;
- whether the shopper belongs to a loyalty program;
- whether the shopper uses a coupon code: and

Benefits of clickstream data analysis

There are a number of benefits organizations can get from clickstream data and clickstream analytics. Among them are the following:

- **User information.** The data collected can include search terms used, pages landed on, webpage features used and the addition or removal of items from a cart, all of which can lead to more actionable insights.
- **User routes.** Organizations can use data analysis to view the different routes their online visitors or customers take to reach a page or to make a purchase.
- **Customer trends and insights.** Collecting and analyzing the clickstreams of a large number of visitors lets an organization identify trends in the following areas:
 - how visitors get to the website;
 - what they do once there;
 - how long they stay on a page;
 - the number of page visits visitors make; and
 - the number of unique and repeat visitors.

- **UX.** If a majority of users quickly leave a page or website, it could be a sign that the page is poorly optimized or doesn't contain enough information of value. Clickstream data enables an organization to [recognize UX shortcomings](#), enabling them to make necessary changes.
- **Digital marketing.** Clickstream data can be used to determine the amount of traffic coming from ad banners and campaigns. Such data provides insight as to which advertisements are most effective and lead to customer [conversion rate optimization](#). Clickstream analysis can also derive what times of day, month or year a marketing strategy is most effective.